



Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CIT 320: Software Development Project-II

Sessional Project Proposal

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Project Title: Bag Shop Inventory

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1. Abstract

Bag Shop Inventory is a Flutter + Supabase based inventory and POS management system tailored for bag retail shops. The system organizes products with bag-specific attributes (type, material, size, barcode), tracks stock per warehouse, records all stock changes as auditable movements (purchase, sale, adjustment, transfer), and supports staff accounts with role-based access. This project focuses on reliability (movement-based inventory), usability for daily shop operations (barcode-first flows), and security (no privileged backend keys embedded in the mobile app; data access controlled through database policies).

2. Objectives

- Build a bag-shop-oriented inventory system with clear product catalog management.
- Implement purchasing workflow (purchase order creation, receiving, and stock update).
- Implement sales/POS workflow (sales order creation, stock deduction, receipt/export support).
- Track stock changes through a stock movement history for accountability.
- Add an Admin route for staff onboarding using invite codes (safe mobile onboarding).
- Provide reporting views for sales and inventory summaries.

3. Problem Statement

Many small retail shops manage inventory manually, causing frequent stock mismatches, slow purchasing/sales processes, and weak accountability. The goal is to create a digital system that maintains accurate, real-time stock and provides a consistent workflow for purchases and sales.

4. Related Work

Existing solutions include general POS and inventory tools such as Shopify POS, Odoo Inventory, Zoho Inventory, and local spreadsheet-based systems. These tools are often either too generic, require paid subscriptions, or are not tailored to the bag shop workflow (barcode-first product lookup, bag-specific attributes, warehouse/batch handling). Bag Shop Inventory aims to provide a

focused, customizable, and academically demonstrable solution using modern cross-platform development and a managed backend.

5. Scope

In-scope

- Authentication (login/register) and company-scoped access.
- Admin panel for staff onboarding via invite codes and staff management.
- Product management with bag-specific fields and barcode support.
- Inventory tracking per warehouse with movement history.
- Purchase order creation and receiving workflow.
- Sales/POS order workflow and stock deduction.
- Reporting screens (initial version: sales summary, stock overview).

Out-of-scope (initial submission)

- Full accounting/tax compliance engine.
- Complex offline-first synchronization (planned as enhancement).
- Advanced demand forecasting / AI prediction.

5.1. Job Market Analysis

Inventory and POS systems are widely demanded in retail businesses, and the skills used in this project are highly relevant in the software job market:

- Cross-platform app development (Flutter/Dart).
- Backend-as-a-service and API integration (Supabase).
- Database modeling and security (PostgreSQL + Row Level Security).
- Clean architecture and state management (Riverpod).
- Real-world workflow design (purchases, sales, stock audit).

Job Portal / Platform	Example Search / Link
BDJobs	https://bdjobs.com/jobs/details/1440397?ln=1
LinkedIn Jobs	https://nextjobz.com.bd/job/IJOB202500392?src=Linkedin
Indeed	https://www.indeed.com/jobs?q=mobile+app&l=&vjk=3860e1d1affd536f

These skills align with roles such as Mobile App Developer, Full-Stack Developer, and Product/Business Application Developer.

6. Methodology

The project will be implemented iteratively:

1. Requirements analysis and database/schema verification.
2. UI and navigation scaffolding.
3. Core workflows: products → purchases → inventory → sales.
4. Reporting and administration features.
5. Testing, bug fixing, and documentation.

6.1. Technology Stack

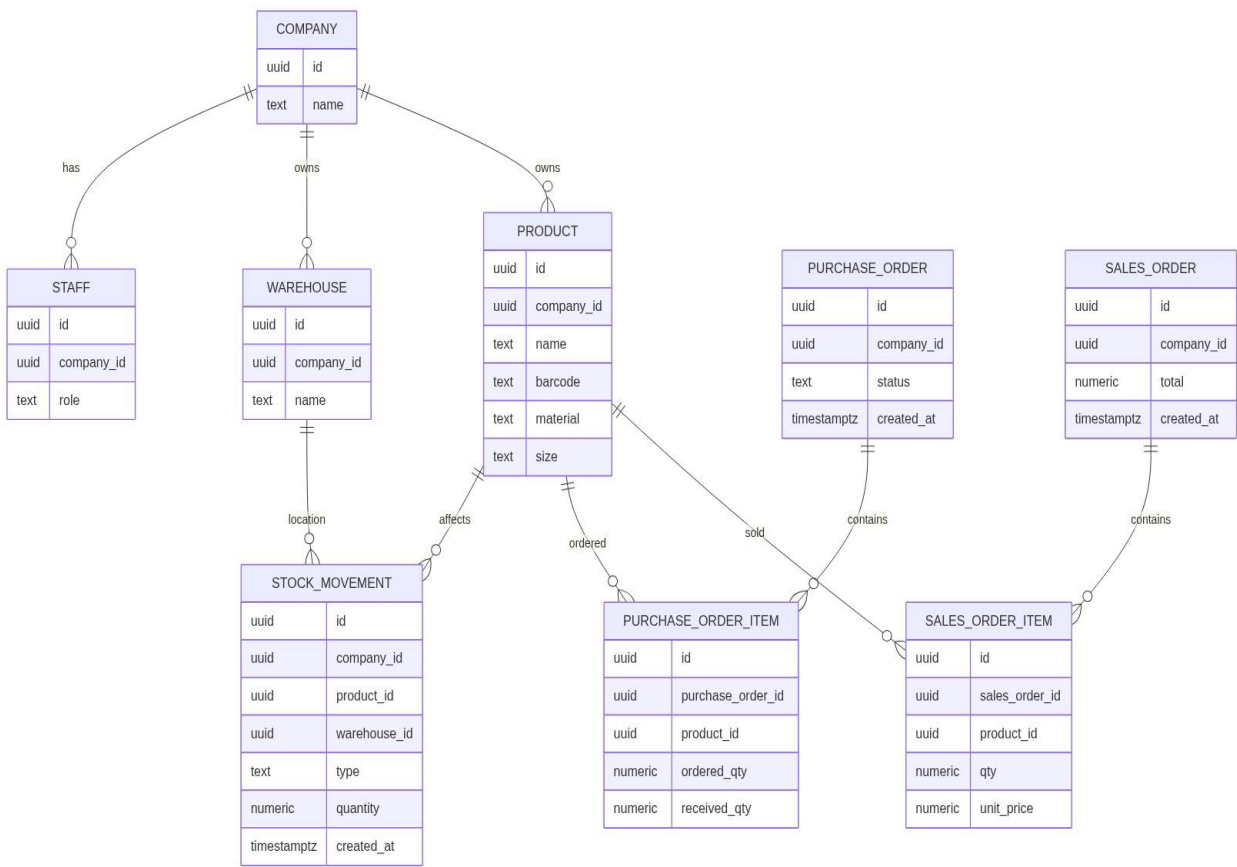
Category	Technology
Frontend	Flutter (Dart), Material UI
State management	Riverpod (flutter_riverpod + hooks_riverpod)
Routing	go_router
Backend	Supabase (PostgreSQL, Auth, Storage)
Barcode scanning	mobile_scanner
Charts / tables	syncfusion_flutter_datagrid, syncfusion_flutter_charts, fl_chart
PDF / printing	pdf, printing
Utilities	intl, uuid, connectivity_plus

6.2. Design Principles

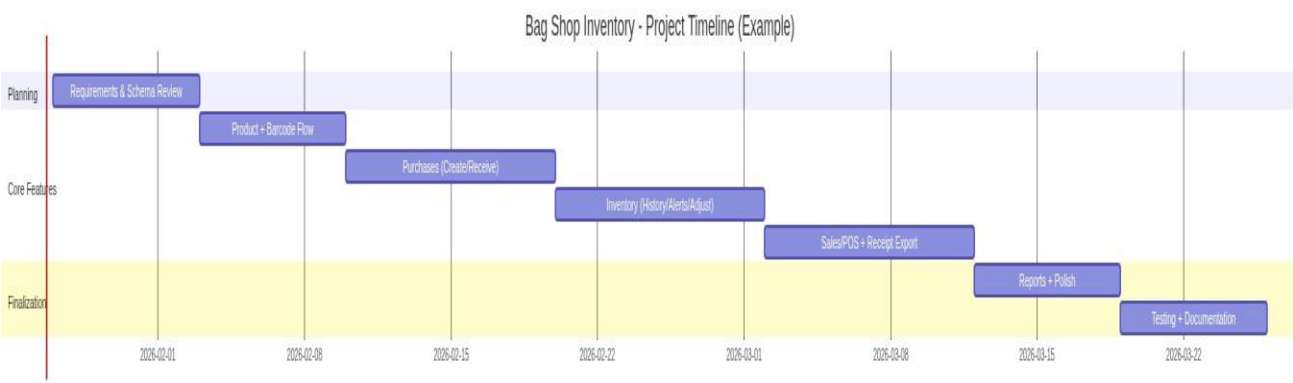
- **Security-first:** no service-role key in client; enforce database Row Level Security.
- **Auditability:** stock changes are stored as stock movement records.
- **Separation of concerns:** presentation/domain/data layers for maintainability.
- **Scalability:** support multiple warehouses and multi-user staff access.
- **User-focused UX:** barcode-first flows and minimal steps for common tasks.

7. Visual Models

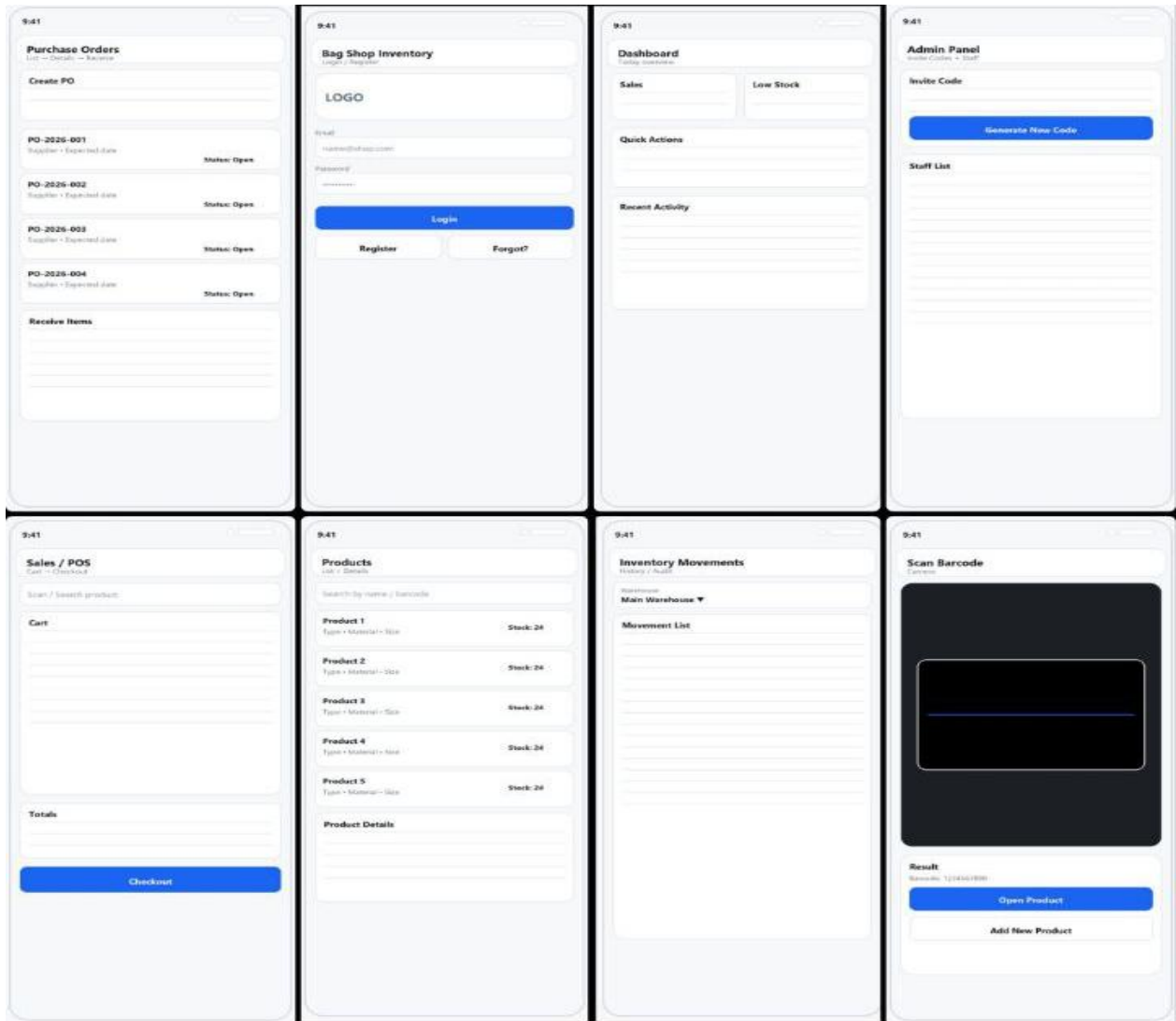
7.1. ERD (Entity Relationship Diagram)



7.2. Timeline (Gantt Chart)



8. UI Mockups



9. Future Plans

- Low-stock alerts and reorder suggestion workflow.
- Stock transfers between warehouses and adjustment reasons.
- Returns/refunds workflow integrated with sales.
- Advanced reports: profit, valuation, supplier analytics.
- Improved offline support and background sync.

10. Result

At the end of the project, the system will deliver:

- A working Flutter application for managing bag shop inventory and POS.

- Secure Supabase backend integration with role-based access.
- Purchase receiving and sales workflows that correctly update stock.
- An auditable stock movement system and initial reporting.

12. Weekly Report

Week	Activities	Output
Week 1	Requirements + Supabase setup + schema/RLS review	Finalized scope and database baseline
Week 2	Product module + barcode integration	Product CRUD and scanning flow
Week 3	Purchase order flow	Create purchase order and list/detail screens
Week 4	Purchase receiving + stock movements	Inventory updates via movements/triggers
Week 5	Sales/POS flow	Sales order creation and stock deduction
Week 6	Reports + UI polish	Sales/stock reports + improved UX
Week 7	Testing + bug fixing	Unit/widget tests and stability improvements
Week 8	Final documentation + demo prep	Submission-ready report and demo